

Boss Challenge — Paper 53 (Class 7)

Motion & Time | Acids, Bases & Salts | Data Handling | Comparing Quantities

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. A scooter covers a distance of 90 kilometres in exactly 3 hours of steady riding. Its speed is:
(A) 30 km/h
(B) 270 km/h
(C) 90 km/h
(D) 3 km/h
2. A drop of lemon juice is placed on moist blue litmus paper. Because lemon juice is acidic, the paper turns:
(A) colourless
(B) green
(C) blue (no change)
(D) red
3. To find the mean (average) of a set of numbers, we add up all the values and then divide the total by:
(A) the largest value
(B) the number of values
(C) two, always
(D) the smallest value
4. A ratio compares two quantities of the same kind by division. The ratio 2 : 5 means that for every 2 units of the first quantity there are:
(A) 3 units of the second
(B) 2 units of the second
(C) 5 units of the second
(D) 10 units of the second
5. When we measure speed as the number of metres travelled for every second of time, the SI unit we are using is the:
(A) kilometre (km)
(B) second (s)
(C) metre per second (m/s)
(D) kilogram (kg)
6. Soap solution is rubbed onto red litmus paper. Being basic, the soap turns the red litmus:
(A) black
(B) red (no change)
(C) blue
(D) yellow

7. The mean (average) of the four numbers 4, 6, 8 and 10 is:
- (A) 6
 - (B) 7
 - (C) 4
 - (D) 28
8. Written as a percentage, the fraction one-quarter ($\frac{1}{4}$) is equal to:
- (A) 14%
 - (B) 25%
 - (C) 4%
 - (D) 40%
9. A toy train that travels the same distance in every equal slice of time is moving with:
- (A) uniform motion
 - (B) circular motion
 - (C) no motion at all
 - (D) non-uniform motion
10. Tamarind, lemon and unripe mango all share a sharp sour taste because each of them contains a(n):
- (A) metal
 - (B) acid
 - (C) salt
 - (D) base
11. Among a list of data values, the one that appears the greatest number of times is called the:
- (A) median
 - (B) mean
 - (C) mode
 - (D) range
12. Half of a number is the same as fifty per cent of it. Therefore 50% of 80 is:
- (A) 50
 - (B) 20
 - (C) 80
 - (D) 40
13. The time taken by a simple pendulum to complete one full to-and-fro swing is called its:
- (A) time period
 - (B) amplitude
 - (C) frequency
 - (D) speed

14. Baking soda solution and soap feel slippery and taste bitter. These properties are typical of substances that are:

- (A) acidic
- (B) neutral
- (C) metallic
- (D) basic (alkaline)

15. When a set of data is arranged in order, the value lying exactly in the middle is called the:

- (A) median
- (B) range
- (C) mean
- (D) mode

16. Expressed as a percentage, the decimal number 0.6 becomes:

- (A) 60%
- (B) 6%
- (C) 0.6%
- (D) 600%

17. Built into a car's dashboard, the instrument that records the total distance the vehicle has travelled is the:

- (A) speedometer
- (B) thermometer
- (C) barometer
- (D) odometer

18. If an acid and a base are combined in just the right amounts, they cancel out to give salt and water. This is called:

- (A) dispersion
- (B) condensation
- (C) neutralisation
- (D) evaporation

19. The range of a set of data is found by subtracting the smallest value from the:

- (A) most frequent value
- (B) largest value
- (C) mean value
- (D) middle value

20. Two cars have speeds of 60 km/h and 40 km/h. Written in simplest form, the ratio of their speeds is:

- (A) 6 : 5
- (B) 2 : 3
- (C) 60 : 40
- (D) 3 : 2

- 21.** On a distance-time graph, a straight slanting line rising steadily to the right represents an object moving with:
- (A) backward motion
 - (B) uniform (constant) speed
 - (C) increasing speed
 - (D) zero speed
- 22.** A few drops of turmeric solution are added to soap water and the yellow turmeric turns red. This colour change shows the soap water is:
- (A) acidic
 - (B) basic
 - (C) neutral
 - (D) salty
- 23.** Over five days the noon temperatures were 20, 22, 24, 26 and 28 degrees Celsius. The mean noon temperature was:
- (A) 24 degrees Celsius
 - (B) 5 degrees Celsius
 - (C) 28 degrees Celsius
 - (D) 120 degrees Celsius
- 24.** If 3 identical notebooks cost 60 rupees, then by the unitary method 5 such notebooks will cost:
- (A) 120 rupees
 - (B) 100 rupees
 - (C) 60 rupees
 - (D) 80 rupees
- 25.** On a distance-time graph, a flat horizontal line over a stretch of time means that during that time the object is:
- (A) speeding up
 - (B) moving at uniform speed
 - (C) at rest (not moving)
 - (D) moving fastest
- 26.** The general name for a substance whose colour change is used to tell whether a liquid is acidic or basic is a(n):
- (A) indicator
 - (B) solvent
 - (C) salt
 - (D) catalyst
- 27.** In a bar graph showing the rainfall of different months, the height of each bar represents the:
- (A) the colour of the bar
 - (B) amount of rainfall (the value of the data)
 - (C) the number of months
 - (D) the name of the month

28. To express any quantity as a percentage, we always write it as a fraction out of:

- (A) 100
- (B) 10
- (C) 50
- (D) 1000

29. A train moves at 36 kilometres per hour. Converted into metres per second, this speed is:

- (A) 36 m/s
- (B) 100 m/s
- (C) 3.6 m/s
- (D) 10 m/s

30. Phenolphthalein is colourless in acids but turns bright pink in a base. When added to lime water it turns:

- (A) colourless
- (B) blue
- (C) red
- (D) pink

31. A car's speed was noted as 30, 40 and 50 km/h at three moments. The mean of these three speed readings is:

- (A) 40 km/h
- (B) 50 km/h
- (C) 3 km/h
- (D) 120 km/h

32. Twenty per cent of the 50 students in a class wear spectacles. The number of students who wear spectacles is:

- (A) 30
- (B) 10
- (C) 50
- (D) 20

33. Over a whole journey a bus covers 150 kilometres in 5 hours, including its halts. Its average speed is:

- (A) 750 km/h
- (B) 30 km/h
- (C) 5 km/h
- (D) 150 km/h

34. Curd and yogurt taste slightly sour because, as milk sets, helpful bacteria produce a substance called:

- (A) common salt
- (B) citric acid
- (C) a base
- (D) lactic acid

35. The mode of the data set 3, 5, 5, 7, 9, 5 and 2 is:

- (A) 7
- (B) 5
- (C) 2
- (D) 9

36. A bottle holds 200 mL of solution, of which 30 mL is acid. The percentage of acid in the solution is:

- (A) 30%
- (B) 6%
- (C) 85%
- (D) 15%

37. After lightning we see the flash almost at once but hear the thunder later. This is because, compared with sound, light travels:

- (A) much slower
- (B) at the same speed
- (C) only as fast as a fast car
- (D) very much faster

38. The acid that gives vinegar its sharp sour smell and taste is:

- (A) citric acid
- (B) hydrochloric acid
- (C) lactic acid
- (D) acetic acid

39. A scientist records the litmus result 'acidic' for 6 samples and 'basic' for 2 samples. The more frequent result - the mode - is:

- (A) acidic
- (B) neutral
- (C) there is no mode
- (D) basic

40. Written as a fraction in its lowest terms, 40% equals:

- (A) $\frac{2}{5}$
- (B) $\frac{4}{10}$
- (C) $\frac{4}{5}$
- (D) $\frac{1}{40}$

41. Car P moves at 60 km/h and car Q at 40 km/h along the same road. The ratio of the speed of P to that of Q is:

- (A) 3 : 1
- (B) 3 : 2
- (C) 60 : 40
- (D) 2 : 3

42. After a heavy meal a person feels acidity in the stomach and takes an antacid tablet. The antacid relieves the discomfort because it is:
- (A) a sugar that coats the stomach
 - (B) an acid that adds more acid
 - (C) a salt that has no effect
 - (D) a base that neutralises the excess acid
43. In a pictograph each picture of a tree stands for 10 trees. A row showing 4 tree pictures therefore represents:
- (A) 10 trees
 - (B) 14 trees
 - (C) 4 trees
 - (D) 40 trees
44. A train covers 200 km in 4 hours. Expressed as a unit rate, its speed is:
- (A) 200 km per hour
 - (B) 4 km per hour
 - (C) 50 km per hour
 - (D) 800 km per hour
45. A cyclist plans to cover 150 km at a steady speed of 50 km/h. The time the journey will take is:
- (A) 3 hours
 - (B) 100 hours
 - (C) 3 minutes
 - (D) 50 hours
46. An ant's sting injects formic acid, causing a burning feeling. Rubbing the spot with baking soda helps because baking soda is:
- (A) neutral, so it does nothing
 - (B) acidic, so it adds more acid
 - (C) a salt that washes it away
 - (D) basic, so it neutralises the acid
47. A fair coin is tossed once. The probability of getting a head is:
- (A) 0
 - (B) $\frac{1}{2}$
 - (C) 1
 - (D) 2
48. A shop offers a 10% discount on a bag marked 500 rupees. The discount amount is:
- (A) 50 rupees
 - (B) 100 rupees
 - (C) 450 rupees
 - (D) 10 rupees

49. A pendulum is timed making 30 complete oscillations in 60 seconds. The time period of the pendulum is:

- (A) 60 seconds
- (B) 2 seconds
- (C) 30 seconds
- (D) 0.5 seconds

50. Common salt (sodium chloride) dissolved in water is tested with both blue and red litmus. The solution turns the litmus:

- (A) red litmus blue, because it is basic
- (B) blue litmus red, because it is acidic
- (C) neither colour - it is neutral
- (D) both papers green

51. Arranged in increasing order, five test marks are 11, 13, 15, 17 and 19. The median (middle) mark is:

- (A) 11
- (B) 15
- (C) 19
- (D) 13

52. Written as a fraction in its simplest form, 75% is equal to:

- (A) $\frac{3}{4}$
- (B) $\frac{7}{5}$
- (C) $\frac{75}{10}$
- (D) $\frac{1}{4}$

53. Two runners start together; after 10 seconds the one who is now farther ahead must have the greater:

- (A) temperature
- (B) speed
- (C) time period
- (D) mass

54. Hydrochloric acid is exactly neutralised by sodium hydroxide. Apart from water, the product formed is the salt:

- (A) vinegar
- (B) more hydrochloric acid
- (C) sugar
- (D) sodium chloride (common salt)

55. A double bar graph (two bars side by side for each category) is most useful when we want to:

- (A) find the area of a shape
- (B) show how one quantity changes over time
- (C) display a single value only
- (D) compare two sets of data side by side

56. When the ratio 15 : 25 is reduced to its lowest terms, it becomes:

- (A) 1 : 2
- (B) 5 : 3
- (C) 3 : 5
- (D) 15 : 25

57. In the SI system, the basic unit used to measure an interval of time is the:

- (A) hour
- (B) minute
- (C) second
- (D) metre

58. A farmer finds a field too acidic for crops. To reduce the acidity, the soil is commonly treated by adding:

- (A) more acid fertiliser
- (B) common salt
- (C) lime (a base)
- (D) vinegar (an acid)

59. For the numbers 12, 15, 20, 9 and 14, the smallest value is 9 and the largest is 20. Their range is:

- (A) 20
- (B) 29
- (C) 9
- (D) 11

60. If 25% of a number is 30, then the whole number is:

- (A) 7.5
- (B) 55
- (C) 120
- (D) 30

61. A bus runs at a steady speed of 60 km/h. In 30 minutes (half an hour) it will cover a distance of:

- (A) 30 km
- (B) 120 km
- (C) 60 km
- (D) 15 km

62. Lime water, used to test for carbon dioxide, turns red litmus blue. This tells us that lime water is:

- (A) basic
- (B) neutral
- (C) acidic
- (D) salty

63. On a summer day a town's highest temperature was 38 degrees and its lowest 22 degrees Celsius. The range of temperature that day was:
- (A) 60 degrees Celsius
 - (B) 22 degrees Celsius
 - (C) 16 degrees Celsius
 - (D) 38 degrees Celsius
64. A toy bought for 200 rupees is sold for 250 rupees. The profit per cent is:
- (A) 50%
 - (B) 20%
 - (C) 25%
 - (D) 25 rupees
65. A car that keeps speeding up, slowing at signals and stopping shows a constantly changing speed. This kind of motion is called:
- (A) non-uniform motion
 - (B) rest
 - (C) uniform motion
 - (D) circular motion
66. Oranges, lemons and other citrus fruits get their tangy sourness mainly from:
- (A) lactic acid
 - (B) acetic acid
 - (C) citric acid
 - (D) a base
67. Before the median of a list of numbers can be read off, the numbers must first be:
- (A) multiplied together
 - (B) counted only
 - (C) arranged in order (ascending or descending)
 - (D) added together
68. In a fruit basket the ratio of apples to the total number of fruits is 1 : 4. The percentage of apples is:
- (A) 4%
 - (B) 25%
 - (C) 20%
 - (D) 75%
69. The needle of a vehicle's speedometer points to a number whose usual unit, printed on the dial, is:
- (A) litres (L)
 - (B) kilometres (km)
 - (C) km/h (kilometres per hour)
 - (D) hours (h)

70. A factory's waste water is found to be strongly acidic. Before releasing it into a river, the factory must first treat it with a base in order to:
- (A) neutralise it
 - (B) make it more acidic
 - (C) colour it blue
 - (D) make it taste sweet
71. A standard six-faced die is rolled once. The probability of getting the number 3 is:
- (A) $\frac{1}{6}$
 - (B) $\frac{3}{6}$
 - (C) 6
 - (D) $\frac{1}{2}$
72. Ten per cent of a quantity is equal to 20. The whole quantity is therefore:
- (A) 2
 - (B) 30
 - (C) 20
 - (D) 200
73. On one distance-time graph two straight lines are drawn for two objects. The steeper line belongs to the object that is moving:
- (A) heavier
 - (B) at rest
 - (C) faster
 - (D) slower
74. A liquid is tested and changes the colour of neither blue litmus nor red litmus. The liquid must therefore be:
- (A) neutral
 - (B) both acidic and basic
 - (C) strongly basic
 - (D) strongly acidic
75. Five readings of mass are 10 g, 20 g, 30 g, 40 g and 50 g. Their mean is:
- (A) 50 g
 - (B) 25 g
 - (C) 150 g
 - (D) 30 g
76. One soap pack has 4 soaps for 80 rupees; another has 6 soaps for 150 rupees. The pack that is the better buy (cheaper per soap) is the:
- (A) pack of 4, at 20 rupees each
 - (B) both cost the same per soap
 - (C) pack of 6, because it has more soaps
 - (D) pack of 6, at 25 rupees each

77. Bike A travels at 50 km/h while bike B travels at 40 km/h. Bike A is faster than bike B by a percentage of:

- (A) 80%
- (B) 20%
- (C) 25%
- (D) 10%

78. Window-cleaning ammonia solution feels soapy and turns turmeric paper red. On the acid-base scale, ammonia solution is:

- (A) acidic
- (B) a salt
- (C) basic
- (D) neutral

79. The probability of an event that is certain to happen, such as the Sun setting today, is:

- (A) 0
- (B) 1
- (C) $\frac{1}{2}$
- (D) 100

80. Out of 20 water samples tested, 8 were found to be acidic. The percentage of acidic samples was:

- (A) 60%
- (B) 20%
- (C) 40%
- (D) 8%

81. A boy walks the 1.5 km from home to school in 30 minutes (half an hour). His walking speed is:

- (A) 30 km/h
- (B) 0.75 km/h
- (C) 3 km/h
- (D) 1.5 km/h

82. In a neutralisation, 30 mL of an acid is exactly cancelled by 30 mL of a base of the same strength. The volume ratio of acid to base used is:

- (A) 1 : 2
- (B) 2 : 1
- (C) 30 : 1
- (D) 1 : 1

83. In a class the most common shoe size - the one worn by the largest number of students - is best described as the:

- (A) mean
- (B) mode
- (C) median
- (D) range

84. Written as a percentage, the fraction $\frac{7}{10}$ equals:

- (A) 7%
- (B) 0.7%
- (C) 17%
- (D) 70%

85. Among a sundial, a sand-clock and a modern quartz wristwatch, the device that keeps time most accurately is the:

- (A) sand-clock
- (B) quartz wristwatch
- (C) sundial
- (D) all are equally accurate

86. Toothpaste is slightly basic. Brushing the teeth with it is helpful chiefly because the toothpaste:

- (A) adds more acid to the teeth
- (B) neutralises the acid made by bacteria in the mouth
- (C) makes the mouth strongly acidic
- (D) has no effect on mouth acid

87. Six students scored 8, 6, 10, 6, 9 and 7 marks. The mode of their scores is:

- (A) 6
- (B) 8
- (C) 9
- (D) 10

88. Two ratios are equivalent when one comes from the other by multiplying both parts by the same number. The ratio equivalent to 1 : 3 is:

- (A) 3 : 1
- (B) 4 : 6
- (C) 2 : 3
- (D) 4 : 12

89. The slope (steepness) of a straight line on a distance-time graph directly tells us the object's:

- (A) speed
- (B) temperature
- (C) starting position
- (D) mass

90. A gardener tests rainwater that has turned slightly acidic. Compared with neutral water, this acidic rainwater will turn blue litmus:

- (A) green
- (B) blue (unchanged)
- (C) white
- (D) red

91. While collecting data, tally marks are used mainly to:

- (A) find the area of a region
- (B) draw accurate circles
- (C) measure the length of objects
- (D) count and record observations quickly

92. A drink is made by mixing sugar and water in the ratio 1 : 4. The percentage of sugar in the whole drink is:

- (A) 20%
- (B) 80%
- (C) 4%
- (D) 25%

93. A motorbike covers 240 km in 4 hours at a steady pace. Its speed is:

- (A) 4 km/h
- (B) 240 km/h
- (C) 60 km/h
- (D) 960 km/h

94. China rose petals soaked in water make a natural indicator that turns dark pink in acids and green in:

- (A) bases
- (B) salts
- (C) pure metals
- (D) acids

95. An event that can never happen - such as drawing a green ball from a bag containing only red balls - has a probability of:

- (A) 1
- (B) 0
- (C) a negative number
- (D) $\frac{1}{2}$

96. The speed of car X is 80 km/h and that of car Y is 100 km/h. The ratio of X's speed to Y's speed, in simplest form, is:

- (A) 80 : 100
- (B) 5 : 4
- (C) 4 : 5
- (D) 8 : 1

97. A pendulum's time period is 2 seconds. The number of complete oscillations it makes in 1 minute is:

- (A) 2
- (B) 30
- (C) 120
- (D) 60

98. Of these four liquids - lemon juice, soap solution, lime water and baking soda solution - the only acidic one is:

- (A) soap solution
- (B) baking soda solution
- (C) lemon juice
- (D) lime water

99. Three numbers have a mean of 8. If a fourth number, 16, is added, the new mean of all four numbers becomes:

- (A) 12
- (B) 10
- (C) 8
- (D) 16

100. A class of 40 children includes 10 left-handed children. The percentage of left-handed children is:

- (A) 10%
- (B) 40%
- (C) 75%
- (D) 25%